

Project Report

📄 AI Business midterm (Both midterm and final concluded)

Content:

v Length: 5-10 pages not including code/table/image.

Member's Info (ID, Major, Name)

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Motivation

According to the WHO, the three major types of cancer are lung, breast, and colorectal cancers, with lung cancer being the most commonly occurring cancer worldwide. As data show, there were approximately 2.5 million new cases, accounting for 12.4% of the total new cases worldwide in 2022. Moreover, lung cancer is also the leading cause of cancer death, with 1.8 million deaths accounting for 18.7% of the total cancer deaths.

Furthermore, not only are people who smoke at risk of getting lung cancer, but so are those who don't smoke. Surprisingly, lung cancer in young adult nonsmokers is on the rise. As the cancer risk keeps increasing, we, being in this demographic group, would like to dive into the topic of lung cancer.

On the other hand, with AI being widely used, applying AI technology in the healthcare system is a promising approach to revolutionize the diagnosis, treatment, and management of various diseases. These days, researchers are working hard on AI medical applications, such as increasing the speed of diagnosis and transcribing spoken language into written words for accurate medical information. Therefore, we found that AI in the medical field might be a crucial topic, and we would like to further study it.

The problem you want to solve

We are interested in knowing what causes lung cancer and why the death rate is so high. What factors other than smoking result in getting the disease? After gaining these insights, what can

we do to prevent lung cancer? How can we communicate the results we find to the public? Last but not least, we seek further medical applications or even business uses.

About the Data:

source, description, exploratory data analysis (EDA),data preprocessing, feature engineering

We downloaded data from Kaggle, but the original dataset had insufficient samples and was somewhat imbalanced. Therefore, we performed data augmentation and balancing to address these issues. Next, proceed with one-hot encoding, converting the original strings into numerical forms to facilitate EDA (Exploratory Data Analysis) and analysis.

Data input: Gender, Age, Children,Smoker, Employed, Years worked, Income level, Social media, Online Gaming

Data output: Cancer

! Smoker、Employed、Social media、Online gaming : 0=no 1=yes

! Gender : 0=male 1=female

! Income level : 0=low 1=medium 2=high

Machine Learning:

model explanation, model selection, model comparison, coding sharing, difficulties

To find out the most accurate model, we selected the SVM/KNN/ANN model for this task, and compared their accuracy rate, confusion matrix, F1 score, to name but a few.

Support Vector Machine (SVM)

A support vector machine (SVM) is a supervised machine learning algorithm that classifies data by finding an optimal line or hyperplane that maximizes the distance between each class in an N-dimensional space. SVMs are commonly used within classification problems and they distinguish between two classes by finding the optimal hyperplane that maximizes the margin between the closest data points of opposite classes. The algorithm, besides, is widely used in machine learning as it can handle both linear and nonlinear classification tasks.

1.Coding Sharing

- Utilized the best parameter to fit the model by GridSearchCV

```

12 # 調参
13 param_grid = {
14     'C': [2400],
15     'gamma': [2.5]
16 }
17
18 grid_search = GridSearchCV(SVC(class_weight='balanced'), param_grid, cv=5)
19 grid_search.fit(X_train, y_train)
20
21 print(f'Best Model Parameters: {grid_search.best_params_}')

```

Best Model Parameters: {'C': 2400, 'gamma': 2.5}

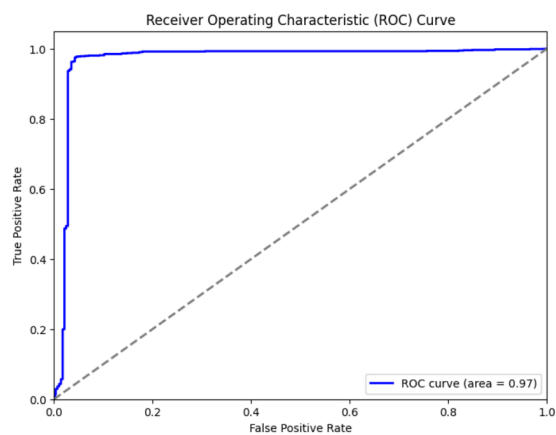
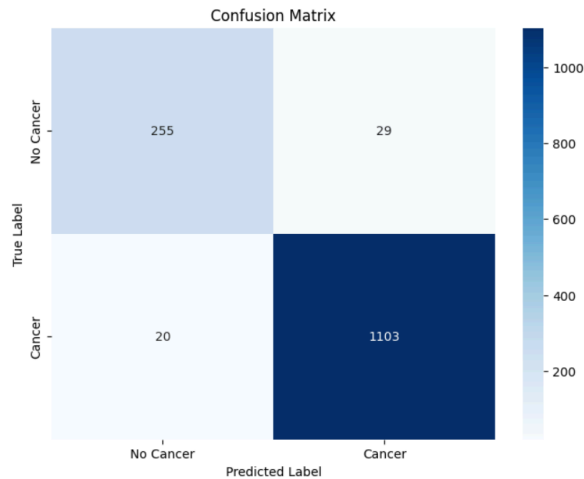
```

1 svm = SVC(**grid_search.best_params_, probability = True)
2 svm.fit(X_train, y_train)
3 y_pred_best = svm.predict(X_test)
4
5
6 # 模型評価
7 accuracy = accuracy_score(y_test, y_pred_best)
8 precision = precision_score(y_test, y_pred_best)
9 recall = recall_score(y_test, y_pred_best)
10 f1 = f1_score(y_test, y_pred_best)
11 conf_matrix = confusion_matrix(y_test, y_pred_best)
12
13 print('Best Model Accuracy: {:.4f}'.format(accuracy))
14 print('Best Model precision: {:.4f}'.format(precision))
15 print('Best Model recall: {:.4f}'.format(recall))
16 print('Best Model f1: {:.4f}'.format(f1))
17 print(f'Confusion Matrix:\n{conf_matrix}')
18

```

Best Model Accuracy:0.9659
 Best Model precision:0.9744
 Best Model recall:0.9831
 Best Model f1:0.9787
 Confusion Matrix:
 [[255 29]
 [19 1104]]

- Visualization of the result : Confusion matrix and ROC



- **Result**
We find the best model accuracy is 0.9652 which is pretty higher than other models.

K-Nearest Neighbour (KNN)

K Nearest Neighbour is a simple algorithm that stores all the available cases and classifies the new data or case based on a similarity measure. It is mostly used to classify a data point based on how its neighbors are classified.

1. Coding Sharing

- Using `StandardScaler()` to transform continuous features

```
[26] #data normalization
continuous_features = ['Age', 'Years Worked']
discrete_features = ['Gender', 'Children', 'Smoker', 'Employed', 'Income Level', 'Social Media', 'Online Gaming']

scaler = StandardScaler()
X_train_continuous = scaler.fit_transform(X_train[continuous_features])
X_test_continuous = scaler.transform(X_test[continuous_features])

X_train_scaled = pd.DataFrame(X_train_continuous, columns=continuous_features)
X_test_scaled = pd.DataFrame(X_test_continuous, columns=continuous_features)

X_train_scaled[discrete_features] = X_train[discrete_features].reset_index(drop=True)
X_test_scaled[discrete_features] = X_test[discrete_features].reset_index(drop=True)

X_train_scaled.head(), X_test_scaled.head()
```

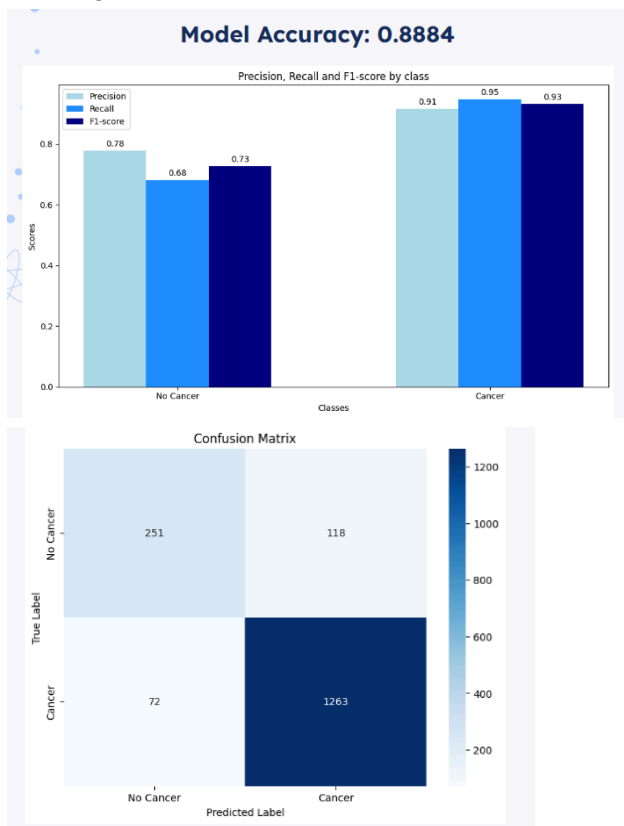
- Fits the model with partially transformed features

```
[27] knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train_scaled, y_train)
```

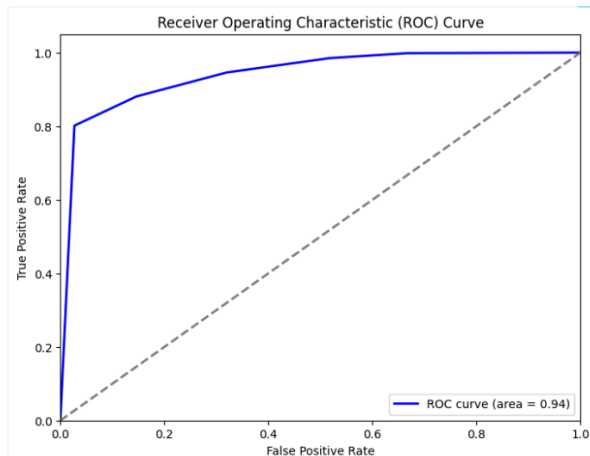
▼ KNeighborsClassifier
KNeighborsClassifier()

```
[28] y_pred = knn.predict(X_test_scaled)
```

- Training Result



- ROC Curve



Artificial Neural Networks (ANN)

An Artificial Neural Network (ANN) is a computational model inspired by the way biological neural networks in the human brain process information. It consists of layers of interconnected nodes (neurons) where each connection represents a weight. ANNs are capable of capturing complex patterns in data and are widely used for tasks such as classification, regression, and even unsupervised learning.

Different from SVM/KNN, ANN is the only deep learning model among the three, which means it requires large amounts of data and higher complexity than other machine learning models. Also, it has more parameters and techniques to deal with.

1. Coding Sharing

- Resemble Data

Since the unbalanced data problem, we used SMOTE to resemble data and make them balanced.

```
from imblearn.over_sampling import SMOTE
import pandas as pd
smote = SMOTE()
X_resampled, y_resampled = smote.fit_resample(X, y)
X_train = X_resampled
y_train = y_resampled
```

- Training Process

We have 9 inputs, and a binary question for classification: Cancer or Not?

```
# Announce a model
mymodel = Sequential()
# i called it "mymodel"
```

```
# Starting to add layer
mymodel.add(Flatten(input_shape=(9,)))
# flatten the matrix image into long vector
```

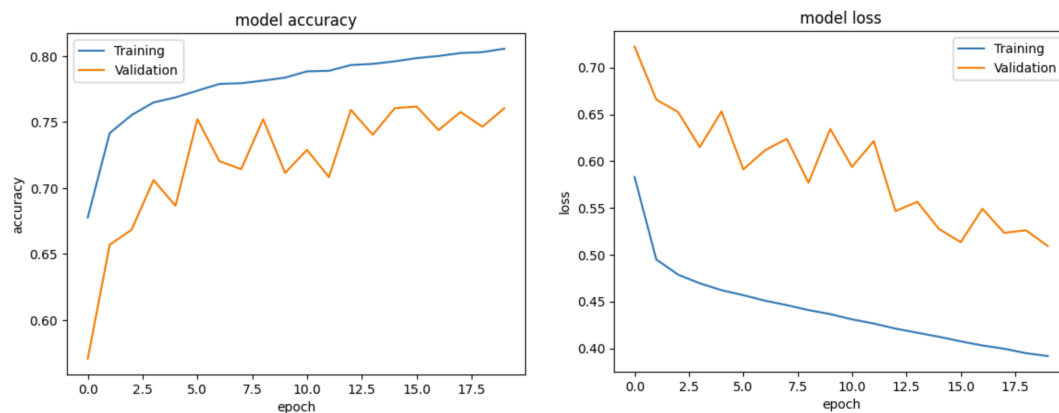
```
# Add 1st hidden layer
mymodel.add(Dense(28, activation='relu'))
```

```
# Add 2nd hidden layer
mymodel.add(Dense(14, activation='relu'))
```

```
mymodel.add(Dense(1, activation='sigmoid'))#只有0或1
# softmax for probability
```

```
mymodel.compile(loss='binary_crossentropy', optimizer='adam', metrics=['accuracy'])
```

• Training Result



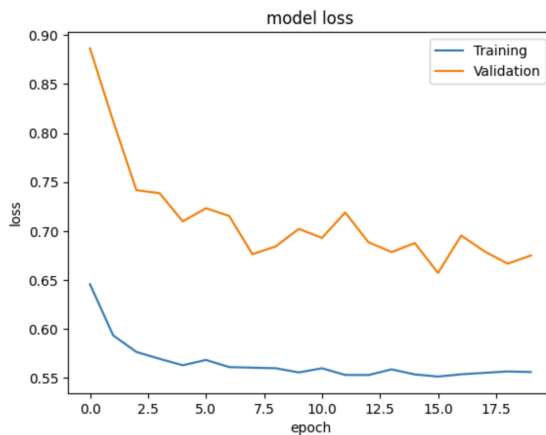
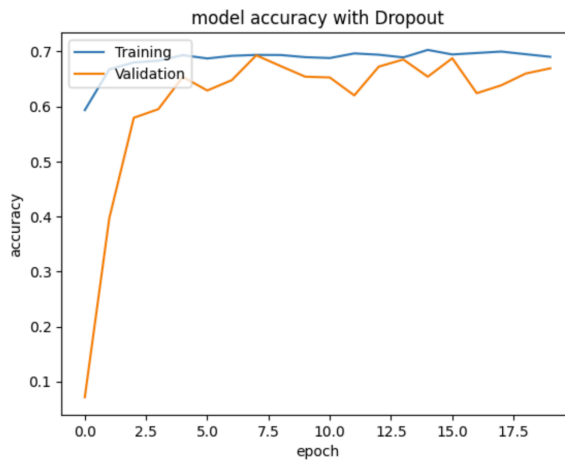
• Dropout

```
# import package for dropout
from keras.layers import Dropout

# Similar step
mymodel_improved = Sequential()
mymodel_improved.add(Flatten(input_shape = (9,)))
#Adding a dropout layer to avoid the overfitting
mymodel_improved.add(Dropout(0.3))
mymodel_improved.add(Dense(28, activation='relu'))
mymodel_improved.add(Dense(14, activation='relu'))
#Adding a dropout layer to avoid the overfitting
mymodel_improved.add(Dropout(0.3))
mymodel_improved.add(Dense(1, activation='sigmoid'))
mymodel_improved.compile(loss='binary_crossentropy', optimizer='adam', metrics=['accuracy'])
```

• Dropout Result

Since there doesn't exist an overfitting problem, dropout show even worse accuracy, or we can try to use a smaller dropout rate in the future.



- Threshold Adjustment

```
# Define predict as the accuracy of the model
fpr, tpr, thresholds = roc_curve(y_test, predict)

# choose optimal value, find the balance between trp and fqe
optimal_idx = np.argmax(tpr - fpr)
optimal_threshold = thresholds[optimal_idx]
```

```
optimal_threshold
```

0.6654142

- Threshold Adjustment Result
Accuracy down to 0.746, the original one is better.
- Softmax activation


```

model = Sequential()
model.add(Dense(64, input_dim=X_train.shape[1], activation='relu'))
model.add(Dense(32, activation='relu'))
model.add(Dense(y.shape[1], activation='softmax'))

model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])
es = EarlyStopping(monitor='val_loss', mode='min', verbose=1, patience=20)

model.fit(X_train, y_train, validation_split=0.1, epochs=100, batch_size=5, callbacks = [es], verbose=1)

y_pred = model.predict(X_test)
y_pred_classes = y_pred.argmax(axis=1)
# y_test_classes = y_test.values.argmax(axis=1)

```

- **Softmax activation**
Quite effective! The accuracy shows to be 0.9133!

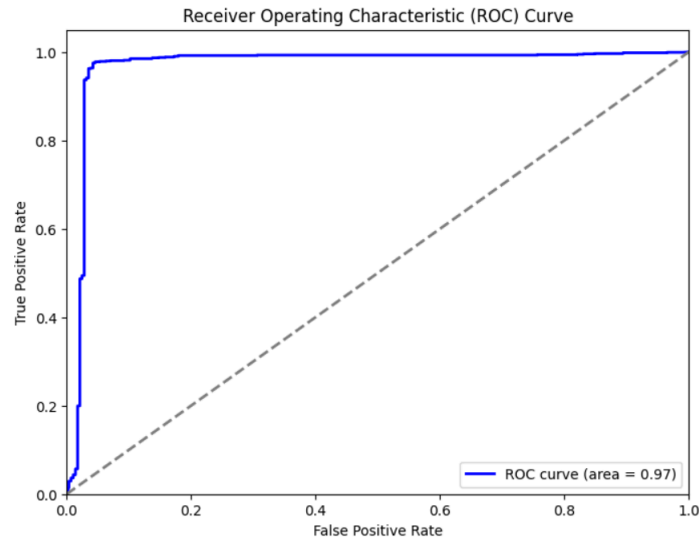
2. Difficulties

We tried lots of methods to improve performances, but the results only show limited improvement. Maybe the reasons are that the dataset isn't big enough for this deep learning model, or the model complexity isn't enough. In future days, we will try more to do our best improvement.

Results

Analysis Insights and Results

Among the three models, SVM, KNN and KNN, SVM has the highest accuracy rate, whose ROC curve is shown in the following graph. Hence, we chose SVM Model as our compass to gain much more insights about the important features that may increase the probabilities of getting lung cancer.



According to the analysis result from SVM Model below, we listed out the Features that have 3 significant impact on getting lung cancer:

	Feature	Importance
0	Gender	-0.000429
1	Age	0.032523
2	Children	0.021078
3	Smoker	0.065451
4	Employed	0.030281
5	Years Worked	0.022651
6	Social Media	0.036218
7	Online Gaming	0.036576
8	Marital_0	0.008798
9	Marital_1	0.000024
10	Marital_2	0.000310
11	Marital_3	-0.000930
12	Income_0	0.003267
13	Income_1	0.016476
14	Income_2	0.009585

1. Smoking habit

- The direct influence between lung disease and nicotine
 - Approximately 80% of lung cancer deaths are due to smoking. Smokers are 15 to 30 times more likely to develop lung cancer compared to non-smokers. The risk increases with the number of cigarettes smoked per day and the duration of smoking

2. Social Media

- Indirect Influences:
 - Mental Health: Excessive use of social media has been associated with various mental health issues such as depression and anxiety. These mental health problems can lead to increased unhealthy behaviors, including smoking, which is a significant risk factor for lung cancer. Social media addiction often results in sedentary lifestyles and neglect of physical health, further contributing to health risks.

- Behavioral Influences: Social media can influence behaviors through peer pressure or exposure to content that promotes smoking or vaping. These platforms can inadvertently normalize smoking, especially among younger users, increasing the likelihood of tobacco use which directly raises lung cancer risk.
- Misinformation and Health Beliefs:
 - Social media is a double-edged sword in terms of health information dissemination. While it can spread awareness, it can also propagate misinformation about lung cancer and its risk factors, leading individuals to underestimate the dangers of smoking and other high-risk behaviors.

3. Online Gaming

- Sedentary Lifestyle:
 - Online gaming is often associated with prolonged sedentary behavior. Spending extensive hours gaming can lead to physical inactivity, which has been linked to various health issues, including an increased risk of lung cancer. Physical inactivity contributes to obesity and other comorbid conditions that can exacerbate cancer risks.
- Mental and Physical Health Impact:
 - Excessive gaming can lead to poor mental health outcomes such as increased stress and anxiety. These mental health issues can further promote smoking or vaping as coping mechanisms, thereby increasing lung cancer risk.
 - There is also a notable overlap between gaming addiction and smoking habits. Many individuals who are heavily involved in gaming might also engage in smoking or vaping, compounding their risk of lung cancer.

What comes to our surprise is that most of us have false stereotypes about the role of gender that will have a certain influence on the probabilities of getting lung cancer. According to the model training outcome, gender didn't have much to do with the result of getting cancer or not.

Business Application

Since we know much more about the plausible causes of getting lung cancer, we have a bunch of ideas that can be tailored into some health software services, healthcare services and even some tangible campaigns to raise the awareness of the public in our daily life, hoping to decrease the probabilities to get lung cancer.

Implementation in Health-Tech Products:

1. **Health Monitoring Platforms:**
 - **Smoking Detection and Alerts:** Integrate machine learning models like SVM into health monitoring apps to identify users with high smoking habits based on their social media and online gaming behaviors. Provide real-time alerts and personalized recommendations for smoking cessation programs.

- **Mental Health Insights:** Use data from social media and online gaming activity to assess mental health states. Deploy interventions for anxiety, depression, and other conditions that could lead to smoking as a coping mechanism.
 - **Activity Tracking:** Incorporate activity monitoring features to detect sedentary behavior associated with excessive gaming and social media use. Suggest physical activities and exercises to mitigate health risks, including those related to lung cancer.
2. **Implementation in Healthcare Services:**
- **Risk Assessment Tools:** Develop risk assessment tools for healthcare providers using SVM models to predict lung cancer risk based on patient lifestyle data. This can help in early detection and personalized treatment plans.
 - **Preventive Programs:** Design targeted preventive healthcare programs focusing on reducing smoking and promoting physical activity. Tailor these programs based on the predicted risk levels from the model.
3. **Implementation in Public Health Campaigns:**
- **Digital Campaigns:** Leverage insights from SVM models to craft effective public health campaigns on social media platforms. These campaigns can address the risks of smoking and sedentary lifestyles, particularly among young adults engaged in online gaming.
 - **Behavioral Nudges:** Use predictive analytics to send behavioral nudges via social media and gaming platforms, encouraging users to adopt healthier habits and reduce their cancer risk.

Conclusion:

A thorough analysis and accurate model training requires much more data and even more features to do better regarding the validation part in the machine learning process. We also consider the potential of Internet of Things (IoT) solutions that may maximize the efficacy of this analysis. However, due to the limitation of ethical and privacy issue, the process and cost of such a concept may be more complicated than we expect. The following are the facets that we propose to make this lung cancer research and analysis excel the most of it:

1. **Model Refinement and Validation:**
- **Data Enrichment:** Collect more diverse and extensive datasets to improve the robustness and generalizability of the SVM model. Include data from various demographics and geographic locations.
 - **Feature Engineering:** Explore additional features that could enhance model performance, such as dietary habits, genetic predispositions, and detailed smoking patterns.

- **Cross-Validation:** Employ cross-validation techniques to further validate the model's performance and ensure its reliability across different datasets.
- 2. **Integration with Emerging Technologies:**
 - **AI and IoT Integration:** Combine machine learning models with Internet of Things (IoT) devices for continuous health monitoring. Wearable devices can provide real-time data on user activities, enhancing the accuracy of health risk predictions.
 - **Natural Language Processing (NLP):** Use NLP to analyze social media content for early detection of health risk behaviors. This can help in identifying users at risk and intervening promptly.
- 3. **Ethical and Privacy Considerations:**
 - **Data Privacy:** Ensure that the collection and use of personal data comply with privacy regulations such as GDPR and HIPAA. Implement robust data encryption and anonymization techniques.
 - **Bias Mitigation:** Continuously monitor and address any biases in the model to ensure fair and equitable healthcare recommendations for all user groups.
- 4. **User Engagement and Education:**
 - **Interactive Tools:** Develop interactive tools and dashboards for users to understand their health risks and the impact of their lifestyle choices. Educate users on how to mitigate these risks effectively.
 - **Feedback Mechanisms:** Incorporate user feedback to refine the predictive models and enhance the overall user experience. Engaged users are more likely to adhere to health recommendations and see better outcomes.

Improvement:

Reflecting suggestions (classmates, TA, & instructor)

Classmates:

1. What is the most difficult part when you collect data ?
The features in the available data are not always ideal, and most of the health issues related data are confidential. So searching for the ideal one wasn't easy, it really took us a while to get this one on Kaggle.
2. What were the main challenges for the coding part ?
Most of us aren't familiar with the difference of models, and the complexity of models. We faced difficulty choosing useful models and improving the accuracy through adjusting parameters. Thanks to everyone's hard work, we finally overcome these challenges.
3. Is the accuracy for the training data or the testing data?
Testing data
4. What is the most difficult part for data preprocessing ?

The most challenging part for us was handling categorical data as converting categorical data into numerical data such as One-Hot encoding can be cumbersome and time-consuming, especially when dealing with a large number of categories.

TA:

According to the problem of SMOTE in SVM, we removed it since the result doesn't change a lot and our final accuracy is 0.9659.

Reference:

WHO "Global cancer burden growing, amidst mounting need for services" [WHO article](#)

BMJ (2022). "Social media and health behaviors." [BMJ Article](#)

American Psychiatric Association. "Technology Addictions: Social Media and More." [APA Article](#)

BMC Public Health (2023). "Impact of Social Media and Gaming on Health." [BMC Public Health Article](#)

Cleveland Clinic. "Physical Inactivity and Cancer Risk." [Cleveland Clinic Article](#)

Science News. "Social Media and Teens' Mental Health." [Science News Article](#)

Coding:

1. SVM: [🔗 AI & Biz_final](#)
2. KNN: Here is link to see the model: [🔗 KNN_model.ipynb](#)
3. ANN: [🔗 AI&Business final ANN不含EDA](#)
4. Data processing: [🔗 Data processing.ipynb](#)